

Network Testing & Validation

1. VLAN Verification

Command:

```
show vlan brief
```

Expected Result

The switch should display:

```
10  ADMIN
```

```
20  IT
```

```
30  GUEST
```

The appropriate switch ports should appear under their respective VLANs.

2. Trunk Verification

Command:

```
show interfaces trunk
```

Expected Result

The router-switch connection should operate as an 802.1Q trunk.

VLANs 10, 20, and 30 should be allowed across the trunk.

3. Router Interface Verification

Command:

```
show ip interface brief
```

Expected interfaces:

```
G0/0.10  192.168.10.1
```

```
G0/0.20  192.168.20.1
```

```
G0/0.30  192.168.30.1
```

The interfaces should be in an operational state.

4. Gateway Testing

Admin

ping 192.168.10.1

Expected:

SUCCESS

IT

ping 192.168.20.1

Expected:

SUCCESS

Guest

ping 192.168.30.1

Expected:

SUCCESS

5. Security Testing

Guest → Admin

ping 192.168.10.10

Expected:

BLOCKED

Guest → IT

ping 192.168.20.10

Expected:

BLOCKED

6. ACL Verification

Command:

show access-lists

The ACL counters should increase when traffic matches the deny rules.

7. Test Summary

Test	Expected Result
Admin → Admin Gateway	PASS
IT → IT Gateway	PASS
Guest → Guest Gateway	PASS
Admin → IT	PASS
IT → Admin	PASS
Guest → Admin	BLOCKED
Guest → IT	BLOCKED
VLAN Configuration	PASS
Trunk Configuration	PASS
ACL Configuration	PASS

Conclusion

Testing confirms that VLAN segmentation, inter-VLAN routing, and ACL-based security policies operate according to the intended design.